

Studio 6

Composition

Objectives

To introduce the concept of composition.

Exercise 1

One object is stored inside another object to store data.

1. What is composition?
2. Download the Studio 6 Ex 1 template project from Moodle. We want to extend the application to store the student's house number, street name, suburb and city separately. We also want to be able to use this format in other sketches if we so desire without having to write the code again. One way to achieve this is to create a class to store information about an address and that class can be easily added to other projects.
3. Create a new tab called Address and create the class definition for the Address class. Create private fields for the information about an address as described above.
4. Create an appropriate constructor for the class.
5. Create a toString() method which returns back the address with the values separated by spaces.
6. In the Student class make the necessary changes to use the Address class to store the address of the student
7. In the sketch code make the necessary changes when creating student objects.
8. Write the code to load a csv file of student data and then display all of the students information to the console window.

Exercise 2

1. Extend the Address class and create get and set methods for all of the fields.
2. In the Student class create a getSuburb() method that returns back the suburb of the student.
3. In the sketch code at the end of the setup method display just the suburb of each student on separate lines.

Exercise 3 (Studio Handin exercise)

1. Create a new sketch and add in the setup and draw methods.
2. You have decided to create your own social media app and make your fortune. You first need to store profile information about people on the app. Create a new class called Profile. It should store the username of the person, the name of the person, their address (*use the class from Ex 1 by copying the file and pasting it into your Studio 6 Ex 3 processing folder*) and their contact details (email address and phone).
3. Create a class called ContactDetails which has private fields to store the email address and phone number as strings.
4. Create an appropriate constructor, get and set methods for all fields and a toString() method.
5. In the Profile class create the private fields required and use an Address object and ContactDetails object to store that information and create the appropriate constructor.
6. Create a toString() method that returns all information about the person including their address and contact details.
7. In the sketch code create an array list to store profiles of people on the app. Then load the profile data from the csv file provided on Moodle and display information from all the profiles to the console window.
8. Make the necessary changes to the classes and sketch code to display just the name and email address of each profile to the console window.

[Use the address class from ex1 for this...](#)